

CSE3190-Fundamentals of Data Analytics

Module3

- Proportion tests
- Chi squared test
- Fisher exact test
- Correlation-T test
- Wilcoxon Rank sum tests
- Wilcoxon signed rank test
- one-way ANOVA test
- Kruskal Wallis test



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Proportion tests

- **Definition:** A proportion test is used to determine whether a sample proportion differs from a hypothesized population proportion.
- **Applications:** Quality control, hypothesis testing, surveys, and reliability testing in engineering.
- **Binary outcomes:** When the variable has two outcomes (e.g., success/failure).



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- **Hypothesis:** Testing a claim about population proportions.
- **Examples:**
 - Quality assurance: Proportion of defective items in a batch.
 - Survey data: Proportion of users preferring a new technology.



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- Key Terms Used

- **Sample Proportion (p):** The proportion of a specific outcome in your sample.
- **Population Proportion (P):** The proportion you are testing against (hypothesized value).
- **Null Hypothesis (H_0):** Assumes no difference between sample proportion and population proportion.
- **Alternative Hypothesis (H_1):** Assumes there is a difference.



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- Types of Proportion Tests
 - **One-proportion Z-test:** Testing a single sample against a known proportion.
 - **Two-proportion Z-test:** Comparing the proportions of two independent samples.



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- Steps in Conducting a Proportion Test

- 1. Formulate Hypotheses:**

- $H_0: p = P$
 - $H_1: p \neq P$ (two-tailed) or $p > P$, $p < P$ (one-tailed)

- 2. Calculate the Test Statistic:**

- Formula:
$$Z = \frac{p - P}{\sqrt{\frac{P(1-P)}{n}}}$$

- Where:

- p = sample proportion
 - P = hypothesized proportion
 - n = sample size



3. Determine Critical Value or P-value

4. Make Decision:

- Reject H_0 if test statistic is greater than critical value or P-value is less than significance level (α).

5. Conclusion: State the result in the context of the problem.



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Example: One-Proportion Z-Test

- **Problem:** A manufacturer claims that only 5% of its products are defective. A quality control engineer inspects 200 products and finds that 12 are defective. Is the manufacturer's claim true?
- **Hypotheses:**
 - $H_0: P = 0.05$
 - $H_1: P \neq 0.05$



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- **Solution:**

- Use Z-test formula to calculate the Z value and compare with the critical value.
 - $P = 0.05$ # population proportion (hypothesized proportion)
 - $n = 200$ # sample size
 - $x = 12$ # number of defective items found in the sample
- Sample proportion $p = \frac{12}{200} = 0.06$

- $$Z = \frac{p - P}{\sqrt{\frac{P(1-P)}{n}}}$$



$$\begin{aligned}
 &= \frac{0.06 - 0.05}{\sqrt{\frac{0.05(1-0.05)}{200}}} = \frac{0.01}{\sqrt{\frac{0.05(0.95)}{200}}} \\
 &= \frac{0.01}{\sqrt{0.0002375}} \\
 &= \frac{0.01}{0.0154110350074} \\
 &= 0.6488856845239
 \end{aligned}$$



- Assuming a significance level of 0.05 for a two-tailed test
 - $\alpha = 0.05$
 - $\text{critical_value} = 1.96$ # Critical value for a 95% confidence interval in a two-tailed test
- The calculated Z value is **0.65**. For a two-tailed test at a 95% confidence level, the critical value is **1.96**.



- Since the Z value (0.65) is less than the critical value (1.96), we fail to reject the null hypothesis. Therefore, there is not enough evidence to conclude that the proportion of defective products differs from the claimed 5%



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Suppose a company claims that **10%** of its products are defective. A quality control engineer checks a random sample of **150** products and finds that **20** of them are defective. The engineer wants to test whether the actual proportion of defective products differs from the company's claim at a **5% significance level**.



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- **Hypotheses:**

- $H_0: P = 0.10$
- $H_1: P \neq 0.10$

- **Solution:**

- Use Z-test formula to calculate the Z value and compare with the critical value.
 - $P = 0.10$ # population proportion (hypothesized proportion)
 - $n = 150$ # sample size
 - $x = 20$ # number of defective items found in the sample
- Sample proportion $p = \frac{20}{150} = 0.13333$



$$\begin{aligned}
 \bullet Z &= \frac{p - P}{\sqrt{\frac{P(1-P)}{n}}} \\
 &= \frac{0.13333 - 0.10}{\sqrt{\frac{0.10(1-0.10)}{150}}} = \frac{0.03333}{\sqrt{\frac{0.10(0.90)}{150}}} \\
 &= \frac{0.03333}{\sqrt{0.0006}} \\
 &= \frac{0.01}{0.02449489742}
 \end{aligned}$$

$$\bullet 1.360827634879543$$



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- For a **two-tailed test** at a **5% significance level** ($\alpha = 0.05$):

The critical value for a 95% confidence interval is **± 1.96** .

- Since the calculated Z-value (**1.36**) is **less than the critical value (1.96)**, we **fail to reject the null hypothesis**. This means there is **not enough evidence** to conclude that the proportion of defective products differs from the company's claim of 10%.

- The test does not provide sufficient evidence to suggest that the actual proportion of defective products is different from 10%



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- **Two-Proportion Z-Test** step by step.
- Suppose two different assembly lines, Line A and Line B, are producing the same product. A quality control manager wants to determine whether there is a significant difference between the proportion of defective products produced by each line.
 - From Line A, a sample of **200** products is inspected, and **30** defective products are found.
 - From Line B, a sample of **250** products is inspected, and **20** defective products are found.



- **Step 1: State the Hypotheses**

- **Null Hypothesis (H_0):** The proportion of defective products is the same for both lines.

$$H_0: P_A = P_B$$

- **Alternative Hypothesis (H_1):** The proportion of defective products differs between the two lines.

$$H_1: P_A \neq P_B$$



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- **Step 2: Collect and Define the Data**
- $n_A=200$ #Sample size from Line A.
- $x_A=30$ #Number of defective products from Line A.
- $n_B=250$ #Sample size from Line B.
- $x_B=20$ #Number of defective products from Line B.



- Sample proportion from Line A (p_A):

$$p_A = \frac{x_A}{n_A} = \frac{30}{200} = 0.15$$

- Sample proportion from Line B (p_B):

$$p_B = \frac{x_B}{n_B} = \frac{20}{250} = 0.08$$



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- **Step 3: Calculate the Test Statistic (Z-Value)**
- The formula for the Z-test for two proportions is:

$$Z = \frac{p_A - p_B}{\sqrt{p(1-p)\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}}$$

- Where:

- p is the pooled sample proportion:

$$p = \frac{X_A + X_B}{n_A + n_B}$$

$$p = 0.111$$

$$Z = 2.347871376374779$$



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- **Step 4: Determine the Critical Value or P-Value**
 - For a **two-tailed test** at a **5% significance level** ($\alpha = 0.05$):
 - The critical value for a 95% confidence interval is ± 1.96 .
- **Decision:** Since the calculated z-value (2.34) is greater than the critical value (1.96), we reject the null hypothesis.
- **Conclusion:** There is sufficient evidence to conclude that there is a significant difference in the proportion of defective products produced by the two assembly lines

Chi-Square Test

- A **Chi-Squared Test** is a statistical test used to determine whether there is a significant association between categorical variables. There are two main types of Chi-Squared Tests:
 - **Chi-Squared Test of Independence:** Used to determine if two categorical variables are independent.
 - **Chi-Squared Goodness-of-Fit Test:** Used to determine if a sample fits a population distribution.



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Example: Chi-Squared Test of Independence

Problem:

A researcher wants to find out if there is an association between **gender** (Male/Female) and **preference for a product** (Like/Dislike). They conduct a survey with the following results:

	Like	Unlike	Total
Male	30	20	50
Female	20	30	50
Total	50	50	100

The researcher wants to test at a 5% **significance level** whether gender is associate with product preference.



- **Step 1: State the Hypotheses**
 - **Null Hypothesis (H_0):** Gender and product preference are independent.
 - **Alternative Hypothesis (H_1):** Gender and product preference are not independent (there is an association).



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- **Step 2: Observed Values (O)**

- The observed values are the counts from the survey:

	Like	Dislike	Total
Male	30	20	50
Female	20	30	50
Total	50	50	100



• Step 3: Calculate Expected Values (E)

- The expected values are calculated using the formula:

- $$E = \frac{(\text{Row Total}) * (\text{Column Total})}{\text{Grand Total}}$$

Let's calculate the expected values for each cell.

	Like	Dislike
Male	25.0	25.0
Female	25.0	25.0



- **Step 4: Calculate the Test Statistic (Chi-Squared Value)**

- The Chi-Squared test statistic is calculated using the formula:

- $\chi^2 = \sum \frac{(O-E)^2}{E}$

- The calculated Chi-Squared test statistic is **4.0**.



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- **Step 5: Determine the Critical Value**

- Degrees of freedom (df) = (number of rows - 1) * (number of columns - 1)
 $df = (2-1)(2-1) = 1$
- At a **5% significance level ($\alpha = 0.05$)**, the critical value for **1 degree of freedom** is **3.841**.



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- **Step 6: Decision**

- Since the Chi-Squared test statistic (**4.0**) is **greater than the critical value (3.841)**, we **reject the null hypothesis**.

- **Conclusion:**

- There is enough evidence to conclude that there is a significant association between gender and product preference.



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Fisher exact test

- The **Fisher Exact Test** is used to determine if there are nonrandom associations between two categorical variables, especially in small sample sizes.
- It's often used as an alternative to the Chi-Squared test when sample sizes are small or when expected frequencies in any of the cells are below 5.



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Example: Fisher's Exact Test

Problem:

Suppose a study is conducted to determine if a new drug is effective in curing a certain disease. The results are summarized as follows:

	Cured	Not Cured	Total
Drug	8	2	10
Placebo	3	7	10
Total	11	9	20

We want to know if there is a significant association between treatment type (Drug or Placebo) and outcome (Cured or Not Cured).



- **Step 1: State the Hypotheses**

- **Null Hypothesis (H_0):** There is no association between the treatment type and the outcome (they are independent).
- **Alternative Hypothesis (H_1):** There is an association between the treatment type and the outcome (they are not independent).



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- **Step 2: Perform Fisher's Exact Test**

- Formula for Fisher's Exact Test:

	Cured	Not Cured	Total
Drug	a	b	(a+b)
Placebo	c	d	(c+d)
Total	(a+c)	(b+d)	n

- The probability of observing this specific arrangement is given by:

$$P = \frac{\binom{a+b}{a} \binom{c+d}{c}}{\binom{n}{a+c}} = \frac{(a+b)!(c+d)!(a+c)!(b+d)!}{n!a!b!c!d!}$$



step 3: Observed Table

The observed table is:

	Cured	Not Cured	Total
Drug	8	2	10
Placebo	3	7	10
Total	11	9	20

Probability of the Observed Table

Using the formula:

The probability of the **observed table** is $P_{\text{observed}}=0.0321$



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Step 4: Adjust Entries to Create More Extreme Tables

We adjust the values in the table while ensuring the marginal totals are maintained.

Step-by-Step Adjustments:

1. Increase the "Cured" count for the Drug group (row)

- 1. If the Drug group cures **9** patients (instead of 8), the "Not Cured" count for the Drug group drops to **1**.
- 2. To maintain the column totals, the Placebo group must cure **2** patients (instead of 3), and **8** will remain "Not Cured".

This gives the table: **More Extreme Table**

- A table more extreme than the observed one (in terms of association between Drug and Cured) is:

	Cured	Not Cured	Total
Drug	9	1	10
Placebo	2	8	10
Total	11	9	20

Step 5: Probability of the Extreme Table

- Using the same formula:
- The probability of the **extreme table** is $P_{\text{extreme}}=0.00268$.

• Step 6: Total p-value

- The total p-value is the sum of the probabilities for the **observed table** and the **extreme table**:
- $p=P_{\text{observed}}+P_{\text{extreme}}$
- $p = p=0.0321+0.00268=0.03478$



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Step 7: Compare with Significance Level

- Significance level (α) = 0.05.
- Total p-value $p=0.03478$ is **less than 0.05**.

Final Conclusion:

Since $p < \alpha$, we **reject the null hypothesis**.

There is sufficient evidence to conclude that there is a statistically significant association between treatment type (Drug or Placebo) and the outcome (Cured or Not Cured) at the 0.05 significance level.



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Correlation-T test

- The **Correlation T-Test** is used to determine if the correlation coefficient (r) between two variables is significantly different from zero.
- In other words, it tests whether there is a significant linear relationship between two continuous variables.



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• Formula for Correlation T-Test Calculation

- The test statistic t for the correlation t-test is calculated using the following formula:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Where:

- r is the sample correlation coefficient.
- n is the number of paired data points (sample size)



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- **Example:**

- Let's say we have a sample of 10 paired data points and the calculated correlation coefficient r between two variables is 0.65. We want to test whether this correlation is significantly different from zero at a 5% significance level.

- **Step 1: Hypotheses**

- $H_0: \rho=0$ (There is no linear relationship between the variables).
- $H_1: \rho \neq 0$ (There is a significant linear relationship between the variables).



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- **Step 2: Given Data**
 - Correlation coefficient (r) = 0.65
 - Sample size (n) = 10
- **Step 3: Apply the correlation t-test formula**
 - The calculated **t-value** is approximately 2.42.
- **Step 4: Determine the Critical Value or P-value**
 - Degrees of freedom $df = n - 2 = 10 - 2 = 8$
 - Using a t-distribution table or calculator, the critical t-value for a **two-tailed test** at a **5% significance level** ($\alpha = 0.05$) and **8 degrees of freedom** is approximately **± 2.306** .



- **Step 5: Decision**

- Since the calculated t-value (2.42) is **greater than the critical value (2.306)**, we **reject the null hypothesis**.

- **Conclusion:**

- There is enough evidence to suggest that the correlation between the two variables is significantly different from zero, indicating a significant linear relationship between them.



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Wilcoxon Rank sum tests

- The **Wilcoxon Rank-Sum Test** (also called the **Mann-Whitney U Test**) is a non-parametric test used to compare whether two independent samples come from the same distribution.
- It's used when the data do not necessarily follow a normal distribution, making it a non-parametric alternative to the two-sample t-test.



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- **Key Points:**

- It tests whether the distributions of two independent samples are the same.
- It is used for ordinal data or continuous data that is not normally distributed.
- Unlike the t-test, it does not assume normality.



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- **Test Procedure:**

1. **Rank the data:** Combine the two samples and rank the values from lowest to highest.
2. **Assign ranks:** Assign ranks to each observation in the combined dataset. If there are ties, assign the average rank to the tied values.
3. **Sum the ranks for each sample:** Compute the sum of ranks for each sample.
4. **Calculate the test statistic:** The test statistic is based on the rank sums, and it compares how the ranks are distributed between the two samples.



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- **Example:**

- Suppose we have two independent samples:
 - **Sample 1:** 3, 1, 7, 5
 - **Sample 2:** 4, 6, 8, 2



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- **Step 1: Combine and Rank the Data**
 - First, combine the two samples and assign ranks.

Value	Rank
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8



- **Step 2: Assign Ranks to Each Sample**

- **Sample 1** (3, 1, 7, 5): Ranks = 3, 1, 7, 5 → Sum of Ranks for Sample 1 = **16 = R1**
- **Sample 2** (4, 6, 8, 2): Ranks = 4, 6, 8, 2 → Sum of Ranks for Sample 2 = **20 = R2**

- **Step 3: Test Statistic**

- For the Wilcoxon Rank-Sum Test, we choose the smaller of the two rank sums:
- $W = \min(R1, R2) = \min(16, 20) = 16$



- **Step 4: Compare to Critical Value or Use Normal Approximation**

- For small sample sizes, you would typically refer to a table of critical values for the Wilcoxon Rank-Sum Test. However, since both samples have 4 observations, we can use a normal approximation.
- To calculate the Z-score using the **Mann-Whitney U statistic**:
 - **Step 4.1: Calculate the U-statistic for Sample 1:**

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1$$

Where

- n_1 and n_2 are the sizes of the two samples.
- R_1 is the rank sum for sample 1.



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- The **Mann-Whitney U-statistic** for Sample 1 is **10**.
- Using the normal approximation, the calculated **Z-score** is approximately **0.577**.
- **Step 5: Decision**
 - At a typical significance level ($\alpha=0.05$), the corresponding critical Z-value for a two-tailed test is **± 1.96** .
 - Since the calculated Z-score (**0.577**) is **less than 1.96**, we **fail to reject the null hypothesis**.
- **Conclusion:**
 - There is no significant evidence to suggest that the two samples come from different distributions. Thus, we conclude that there is no significant difference between the two groups based on the Wilcoxon Rank-Sum Test.



Wilcoxon signed rank test

- The **Wilcoxon Signed-Rank Test** is a non-parametric test used to compare two related samples or paired observations to assess whether their population mean ranks differ.
- It is the non-parametric equivalent of the paired t-test and is used when the data does not follow a normal distribution.



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- **When to Use:**

- **Paired samples:** When you have two sets of related data, such as pre-test and post-test measurements from the same subjects.
- The differences between the paired observations are used for ranking.



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- **Test Procedure:**

1. **Calculate the differences** between each pair of observations.

2. **Rank the absolute values** of the differences from smallest to largest. Assign the average rank to ties.

3. Assign the **signs** (+ or –) of the differences to the ranks.

4. **Sum the ranks** for the positive differences (R_+) and the negative differences (R_-).

5. The test statistic W is the smaller of R_+ and R_-



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• Example

- Suppose we have the following paired data representing the weights of individuals before and after a fitness program:

Person	Before(kg)	After(kg)
1	70	68
2	82	80
3	76	75
4	85	85
5	74	72
6	90	88



- **Step 1: Calculate the difference**

Person	Before(kg)	After(kg)	Difference (After-Before)
1	70	68	-2
2	82	80	-2
3	76	75	-1
4	85	85	0
5	74	72	-2
6	90	88	-2



• Step 2: Rank the Absolute Differences:

- Ignore differences of zero (like for person 4) and rank the absolute differences.
- If there are ties (i.e., two or more differences with the same value), assign the **average rank** to those tied values.

Absolute Difference	Rank
1	1
2	2,2,2,2 Average Rank = $(2+3+4+5)/4$ = $14/4=3.5$

Person	Difference (After-Before)	Absolute Difference	Rank
1	-2	2	3.5
2	-2	2	3.5
3	-1	1	1
4	0	0	Excluded
5	-2	2	3.5
6	-2	2	3.5



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• Step 4: Assign Signs to the Ranks

- Assign the sign of the original difference to the corresponding rank.
 - If the original difference was negative, assign a negative sign to the rank.
 - If the original difference was positive, assign a positive sign to the rank.

Person	Difference (After-Before)	Rank	Signed Rank
1	-2	3.5	-3.5
2	-2	3.5	-3.5
3	-1	1	-1
4	0	Excluded	Excluded
5	-2	3.5	-3.5
6	-2	3.5	-3.5



- **Step 5: Calculate the Sum of the Positive and Negative Ranks**
 - **Sum of Positive Ranks (R_+):**
 - Since all differences are negative, there are no positive ranks, so $R_+ = 0$.
 - **Sum of Negative Ranks (R_-):**
 - Add up the negative ranks:
 - $R_- = 3.5 + 3.5 + 1 + 3.5 + 3.5 = -15$
 - The test statistic W is the smaller of R_+ and R_- , which is:
 - $W = \min(0, 15) = 0$
- **Step 6: Decision**
 - At a typical significance level ($\alpha = 0.05$):
- Example for $n=5, W=0$:
- In the table for $n=5$:
 - $W=0$ corresponds to a two-tailed p-value of 0.03125 per tail.
 - For two-sided tests: Multiply by 2 $\rightarrow p = 2 \times 0.03125$

Conclusion for the Wilcoxon Signed-Rank Test:

Based on the data provided and the results:

- **Test Statistic (W):** The calculated Wilcoxon test statistic is $W=0$
- **p-value:** The **two-tailed p-value** is 0.0625
- **Significance Level (α):** Assuming a significance level of $\alpha=0.05$, we compare the p-value to α .

Decision: Since the p-value (0.0625) is **greater than** the significance level (0.05), we **fail to reject the null hypothesis**.

Conclusion: This means there is insufficient evidence to conclude a significant difference in weights before and after the fitness program.



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Wilcoxon Signed-Rank Table (Two-Tailed p-Values)

n (Non-zero Pairs)	$W = 0$	$W = 1$	$W = 2$	$W = 3$	$W = 4$	$W = 5$
3	0.125	0.250	0.500	-	-	-
4	0.0625	0.1875	0.375	0.625	-	-
5	0.03125	0.125	0.28125	0.5	0.71875	0.875
6	0.015625	0.078125	0.210938	0.421875	0.632813	0.8125
7	0.0078125	0.0546875	0.171875	0.351563	0.578125	0.796875



one-way ANOVA test

- The **One-Way ANOVA** test is used to determine whether there are statistically significant differences between the means of three or more independent (unrelated) groups.
- It is a parametric test used when the assumptions of normality and homogeneity of variance are met.



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- **When to Use:**

- You have one independent variable with three or more levels (groups).
- You want to test if the means of these groups are significantly different.



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- **How One-Way ANOVA Works:**

- 1. Between-group variability:** The variability between the means of the groups.

- 2. Within-group variability:** The variability of observations within each group.

- One-Way ANOVA compares the variance between the groups (due to differences in group means) and the variance within the groups (due to individual differences).



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- **Formula for F-statistic:**

- $F = \frac{\text{Mean Square Between Groups (MSB)}}{\text{Mean Square Within Groups (MSW)}}$

Where:

- **MSB** (Mean Square Between Groups) is the variance between the group means.
 - **MSW** (Mean Square Within Groups) is the variance within the groups.



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- **Steps in a One-Way ANOVA:**
 - **Step 1: Calculate Group Means and Grand Mean**
 - Calculate the mean for each group.
 - Calculate the overall mean (grand mean) across all observations.
 - **Step 2: Calculate the Sum of Squares**
 - **Sum of Squares Between (SSB):** This measures the variability between group means.

$$SSB = \sum_{i=1}^k n_i (\bar{X}_i - \bar{X}_{\text{grand}})^2$$

Where

- n_i is the number of observations in group i ,
- $\bar{X}_i$ is the mean of group i , and
- $\bar{X}_{\text{grand}}$ is the grand mean.



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- **Sum of Squares Within (SSW):** This measures the variability within each group.

$$SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2$$

Where

- X_{ij} is the individual observation in group i
- $\bar{X}_i$ is the mean of group i , and
- **Total Sum of Squares (SST):** The total variability in the data.

$$SST = SSB + SSW$$



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- **Step 3: Calculate the Mean Squares**

- **Mean Square Between (MSB):** Divide the sum of squares between by the degrees of freedom between groups.

$$MSB = \frac{SSB}{k - 1}$$

Where k is the number of groups.

- **Mean Square Within (MSW):** Divide the sum of squares within by the degrees of freedom within groups.

$$MSW = \frac{SSW}{N - k}$$

Where N is the total number of observations.



- **Step 4: Calculate the F-statistic**

$$F = \frac{MSB}{MSW}$$

- **Step 5: Compare the F-statistic to the critical value or calculate the p-value**

- Use an F-distribution table to compare the calculated F-statistic to a critical value, or calculate the p-value.
- If the p-value is less than the significance level (e.g., 0.05), reject the null hypothesis.



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- **Example:**

Suppose we have three different teaching methods, and we want to test if the average scores of students taught using these methods are significantly different. The data are:

- **Group 1 (Method A):** [85, 90, 88]
- **Group 2 (Method B):** [78, 82, 84]
- **Group 3 (Method C):** [92, 94, 89]

- **Null Hypothesis (H_0):**

$$H_0: \bar{X}_1 = \bar{X}_2 = \bar{X}_3$$

- **Alternative Hypothesis (H_1):**

$$H_1: \bar{X}_1 \neq \bar{X}_2 \neq \bar{X}_3$$



• Step 1: Calculate Group Means and the Grand Mean

• Mean of Group 1:

$$\bar{X}_1 = \frac{85 + 90 + 88}{3} = 87.67$$

• Mean of Group 2:

$$\bar{X}_2 = \frac{78 + 82 + 84}{3} = 81.33$$

• Mean of Group 3:

$$\bar{X}_3 = \frac{92 + 94 + 89}{3} = 91.67$$

• Grand Mean

$$\bar{X}_{grand} = \frac{85 + 90 + 88 + 78 + 82 + 84 + 92 + 94 + 89}{9} = 86.89$$



- **Step 2: Calculate the Sum of Squares**
 - **2.1: Sum of Squares Between (SSB)**
 - For Group 1:
 $n_1=3,$
 $(\bar{X}_1 - \bar{X}_{grand})^2 = (87.67 - 86.89)^2 = 0.608$
Contribution to SSB: $3 \times 0.608 = 1.8243$
 - For Group 2:
 $n_2=3,$
 $(\bar{X}_2 - \bar{X}_{grand})^2 = (81.33 - 86.89)^2 = 30.878$
Contribution to SSB: $3 \times 30.878 = 92.634$
 - For Group 3:
 $n_3=3,$
 $(\bar{X}_3 - \bar{X}_{grand})^2 = (91.67 - 86.89)^2 = 22.878$
Contribution to SSB: $3 \times 22.878 = 68.634$



- Now, sum them up to get the total SSB:
 - $SSB = 1.824 + 92.634 + 68.634 = 163.092$



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- **2.2: Sum of Squares Within (SSW)**

- For Group 1 (Method A):
 $(85-87.67)^2+(90-87.67)^2+(88-87.67)^2= 12.6667$
- For Group 2 (Method B):
 $(78-81.33)^2+(82-81.33)^2+(84-81.33)^2=18.667$
- For Group 3 (Method C):
 $(92-91.67)^2+(94-91.67)^2+(89-91.67)^2=12.667$
- Now, sum them up to get the total SSW:
 $SSW= 12.67+18.67+12.67=44.001$



- **Step 3: Calculate the Mean Squares**

- **3.1: Mean Square Between (MSB)**

$$MSB = \frac{SSB}{k - 1} = \frac{163.092}{3 - 1} = \frac{163.092}{2} = 81.546$$

- **3.2: Mean Square Within (MSW)**

$$MSW = \frac{SSW}{N - k} = \frac{31.11}{9 - 3} = \frac{44.001}{6} = 7.335$$



- **Step 4: Calculate the F-statistic**

$$F = \frac{MSB}{MSW} = \frac{81.546}{7.335} = \mathbf{11.117}$$

- **Step 5: Determine Critical Value or p-value**

Compare to Critical Value or Calculate p-value

- **Degrees of Freedom:**

- $df_{\text{between}} = 2(k-1)$

- $df_{\text{within}} = 6(N-k)$



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- **Compare F-Statistic to Critical Value**
- Using an F-distribution table at $\alpha=0.05$:
- The critical F-value for $df_{\text{between}}=2$ and $df_{\text{within}}=6$ is **5.14**.

Since:

- $F=11.11 > F_{\text{critical}}=5.14$
- **Step 6: Compare to Critical Value or Calculate p-value**

Conclusion:

- Since, $F=11.11 > F_{\text{critical}}=5.14$ and The p-value is **0.0096** (less than 0.05). The null hypothesis is rejected, meaning there is a significant difference between the group means.



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- **F-distribution critical value table** for $\alpha=0.05$ significance level:

$df_{between}$	$df_{within} = 3$	$df_{within} = 4$	$df_{within} = 5$	$df_{within} = 6$	$df_{within} = 7$
1	10.13	7.71	6.61	5.99	5.59
2	9.55	6.94	5.79	5.14	4.74
3	8.85	6.59	5.41	4.76	4.35
4	8.45	6.39	5.19	4.53	4.12
5	8.02	6.06	4.95	4.36	3.95

How to Use the Table:

Degrees of Freedom Between Groups ($df_{between}$): Rows of the table.

Degrees of Freedom Within Groups (df_{within}): Columns of the table.

For your case:

$df_{between}=2$

$df_{within}=6$

From the table: The critical value is $F_{critical}=5.14$.



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Kruskal Wallis test

- The **Kruskal-Wallis Test** is a non-parametric alternative to the One-Way ANOVA.
- It is used to determine if there are statistically significant differences between the distributions of three or more independent groups.
- Unlike ANOVA, it does not assume a normal distribution and is based on **ranked data**.



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- **When to Use:**

- You have three or more independent groups.
- You want to test whether the distributions of the groups differ.
- You do not assume the data to be normally distributed.



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- **Test Statistic:**

- The Kruskal-Wallis test uses ranks rather than raw data to compute a test statistic. The formula for the test statistic H is:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1)$$

Where:

- N is the total number of observations across all groups.
- k is the number of groups.
- R_i is the sum of ranks for group i.
- n_i is the number of observations in group i.
- If the null hypothesis is true, H approximately follows a **chi-squared distribution** with k-1 degrees of freedom.



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- **Steps for Performing the Kruskal-Wallis Test:**

- **Step 1: Rank the Data**

- Combine all the data from all groups and assign ranks, with the smallest value getting a rank of 1. In the case of ties, assign average ranks.

- **Step 2: Calculate the Test Statistic H**

- Use the formula in the previous slide to calculate the Kruskal-Wallis test statistic.

- **Step 3: Determine the p-value**

- Compare the calculated H statistic to the critical value from the chi-squared distribution with $k-1$ degrees of freedom, or calculate the p-value.



- **Example:**

Let's use the same data as the previous One-Way ANOVA example, but this time apply the Kruskal-Wallis test:

- **Group 1 (Method A):** [85, 90, 88]
- **Group 2 (Method B):** [78, 82, 84]
- **Group 3 (Method C):** [92, 94, 89]

Calculate the Kruskal-Wallis test statistic and p-value for this example.



- **Step 1: Combine and Rank the Data**

- First, we combine all the data from the three groups and rank the values, assigning the smallest value the rank of 1. In the case of ties, we assign the average rank to the tied values.

- **Combined Data (Unranked):**

- Group 1 (Method A): 85, 90, 88
- Group 2 (Method B): 78, 82, 84
- Group 3 (Method C): 92, 94, 89



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- Combine and Rank:

Score	Rank
78	1
82	2
84	3
85	4
88	5
89	6
90	7
92	8
94	9



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- **Step 2: Calculate the Rank Sums for Each Group**

- Now, assign the ranks back to the original groups and sum them for each group:

- **Group 1 (Method A):** Scores = 85, 90, 88 → Ranks = 4, 7, 5

$$R_1 = 4 + 7 + 5 = 16$$

- **Group 2 (Method B):** Scores = 78, 82, 84 → Ranks = 1, 2, 3

$$R_2 = 1 + 2 + 3 = 6$$

- **Group 3 (Method C):** Scores = 92, 94, 89 → Ranks = 8, 9, 6

$$R_3 = 8 + 9 + 6 = 23$$



- **Step 3: Calculate the Kruskal-Wallis Test Statistic H**
 - **Step 3.1: Calculate Total Sample Size N**
 - N=9(since there are 3 scores in each group and 3 groups)
 - **Step 3.2: Calculate $\frac{R_i^2}{n_i}$ for Each Group**
 - For **Group 1** (Method A):
$$\frac{R_1^2}{n_1} = \frac{16^2}{3} = \frac{256}{3} = 85.33$$
 - For **Group 2** (Method B):
$$\frac{R_2^2}{n_2} = \frac{6^2}{3} = \frac{36}{3} = 12$$
 - For **Group 3** (Method C):
$$\frac{R_3^2}{n_3} = \frac{232}{3} = \frac{529}{3} = 176.33$$



- **Step 3.3: Sum the Terms**

- $\sum_{i=1}^k \frac{R_i^2}{n_i} = 85.33 + 12 + 176.33 = 273.67$

- **Step 3.4: Substitute into the Formula for HHH**

- $$H = \frac{12}{9(9+1)} * 273.67 - 3(9 + 1)$$
$$= 6.49$$



• Step 4: Determine the p-value

- The Kruskal-Wallis test statistic $H=6.49$ approximately follows a **chi-squared distribution** with $k-1=3-1=2$ degrees of freedom.
- Using statistical software or a chi-squared table, we find the **p-value** for $H=6.49$ and 2 degrees of freedom is approximately **0.039**.
- Critical Value ($\chi^2_{0.05,2}$): **5.99**

• Conclusion:

- Since the test statistic $H=6.4$ exceeds the critical value 5.99, or equivalently, since the p-value (0.039) is less than the significance level ($\alpha=0.05$), **we reject the null hypothesis** and conclude that there is a significant difference in the distributions of the three groups.



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- Here is a detailed chi-squared (χ^2) table for degrees of freedom (df) ranging from 1 to 10 at common significance levels (α):

df	0.10 (10%)	0.05 (5%)	0.025 (2.5%)	0.01 (1%)	0.005 (0.5%)	0.001 (0.1%)
1	2.71	3.84	5.02	6.63	7.88	10.83
2	4.61	5.99	7.38	9.21	10.60	13.82
3	6.25	7.81	9.35	11.34	12.84	16.27
4	7.78	9.49	11.14	13.28	14.86	18.47
5	9.24	11.07	12.83	15.09	16.75	20.52
6	10.64	12.59	14.45	16.81	18.55	22.46
7	12.02	14.07	16.01	18.48	20.28	24.32
8	13.36	15.51	17.53	20.09	21.96	26.12
9	14.68	16.92	19.02	21.67	23.59	27.88
10	15.99	18.31	20.48	23.21	25.19	29.59

Thank
you!



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